Imaadh Usoof

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Education

University of Melbourne — Melbourne, AUS
Bachelor of Science, major in Computing and Software Systems — *Jul. 2024 – Present*

Projects

Fitness Tracker App | *Python, FastAPI, PostgreSQL, SQLAlchemy, React, Vite* — January 2026 – Present

- Built a full-stack fitness tracking app combining calorie/food logging and workout tracking, with a FastAPI REST backend, PostgreSQL database, and React frontend
- Integrated the USDA FoodData Central API for nutritional data lookup, and implemented a TDEE-based insights engine to provide personalised calorie and macro targets
- Designed and implemented JWT-based authentication with bcrypt password hashing, and deployed the application to Railway with a CI-ready project structure

PDF annotation app | *Electron, React, TypeScript, SQLite* — March 2025 – June 2025

- Built an application for lecture note taking and PDF annotation using Electron and TypeScript
- Implemented stylus and mouse-based annotation tools, used SQLite to store offline edits
- Ensured app has crucial PDF editing features only found on paid alternatives such as Adobe Acrobat

Space shooter | *Python, pygame, TensorFlow* — May 2019 – August 2019

- Built an endless space shooter game in Python using pygame, featuring asteroid waves, boss battles, and WASD controls
- Trained a reinforcement learning agent using TensorFlow to autonomously play the game, designing the game's control scheme (left/right only) to make it learnable by a simple neural network

Spellchecker Dictionary | *C, Patricia Trie* | Coursework — September 2025 – October 2025

- Implemented a spellchecker in C using a Patricia trie (compressed radix tree), supporting exact and fuzzy key lookup across a large address dataset
- Engineered bit-level string operations including custom getBit and createStem functions to navigate the trie using binary representations of characters
- Applied edit distance algorithms for closest-match recommendations when no exact key is found, traversing all descendant nodes at the mismatch point
- Conducted a formal complexity analysis comparing the Patricia trie against a linked list implementation, varying input size and key characteristics to validate Big-O behaviour empirically

Technical Skills

Languages: Java, Python, C, SQL, JavaScript
Frameworks: FastAPI, React, Electron, SQLAlchemy
Databases: MySQL, PostgreSQL, Amazon RDS
Libraries: Tensorflow, pandas, NumPy, Matplotlib, pygame, PyTorch

Portfolio: imaadhusoof.com